Week 2 Practice Problem Solutions

4. Masked Phone Number Formatter

Validate exactly ten numeric digits and display six X characters followed by a hyphen and the final four digits.

Java Solution

```
public class Main {
    static String maskPhoneNumber(String phone) {
        if (phone.length() != 10) {
            return "Invalid phone number";
        }
        for (int i = 0; i < phone.length(); i++) {
            if (!Character.isDigit(phone.charAt(i))) {
                return "Invalid phone number";
            }
        }

        StringBuilder masked = new StringBuilder("XXXXXX");
        masked.append("-").append(phone.substring(6));
        return masked.toString();
    }

    public static void main(String[] args) {
        System.out.println(maskPhoneNumber("9876543210"));
        System.out.println(maskPhoneNumber("98765"));
    }
}
```

Sample Output

```
XXXXXX-3210
Invalid phone number
```

Notes

The loop checks every character so a ten-character string containing letters is also rejected.